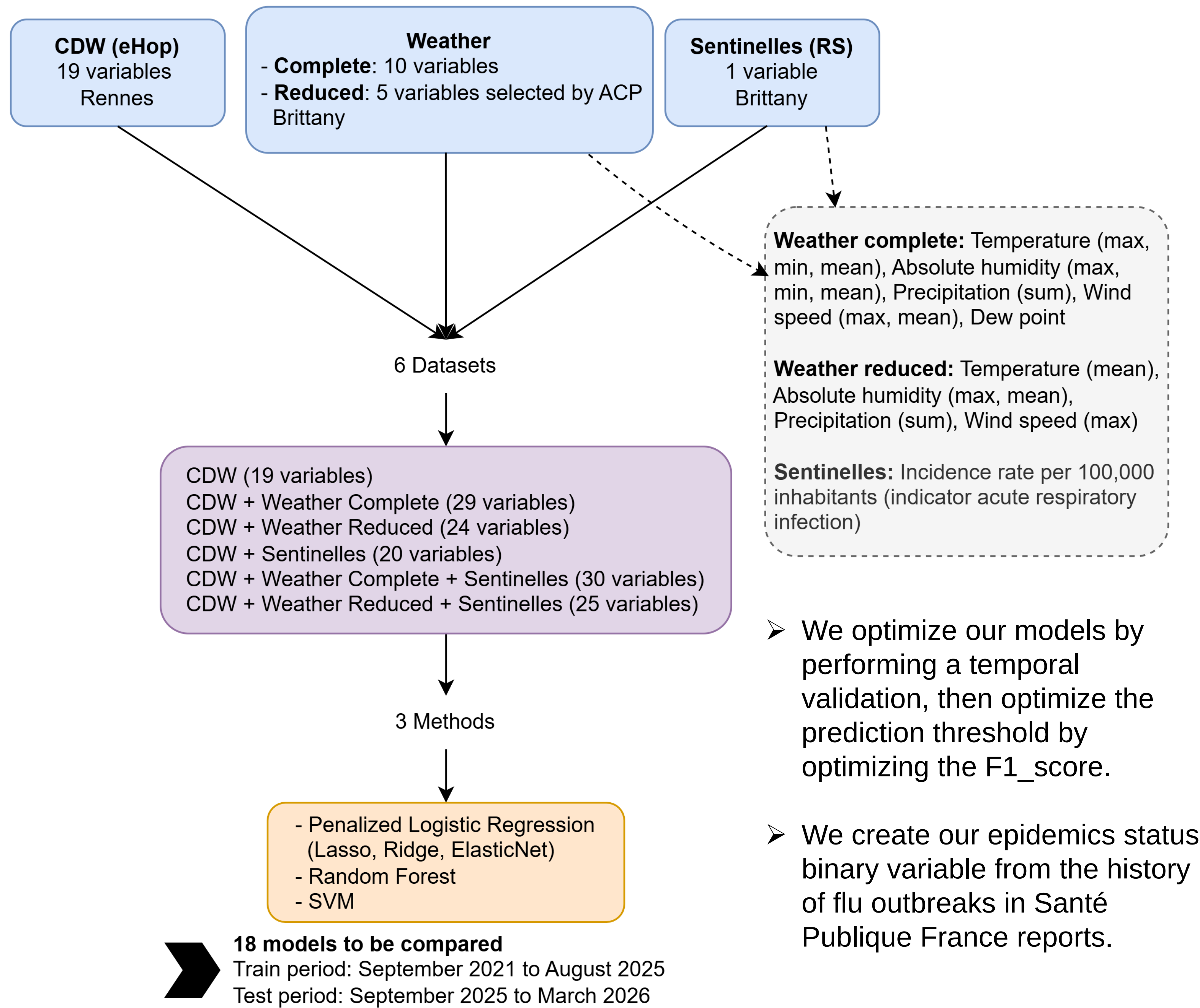


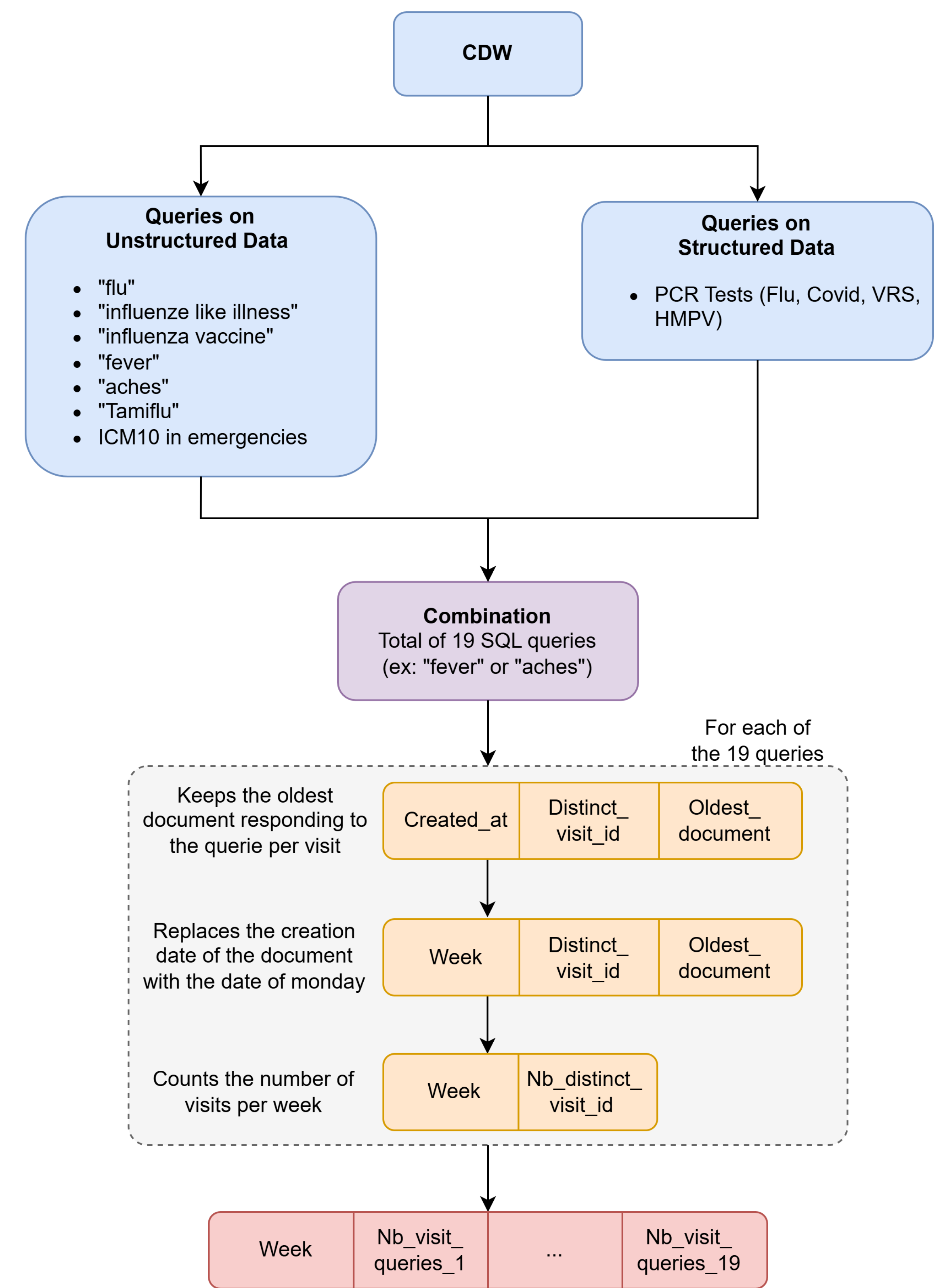
Introduction

- Each year, one billion people are affected by influenza worldwide, causing between 290,000 and 650,000 deaths.
- In France, traditional flu surveillance relies on the Sentinelles system, which provides valuable but delayed indicators (1-3 weeks).
- Clinical Data Warehouses (CDWs) can reduce this lag by using structured and unstructured hospital data to deliver timely, validated case counts enriched with clinical details.
- The objective is to improve responsiveness to outbreaks by exploiting CDWs combined with heterogeneous sources, in order to predict the epidemic status of the following week, with a particular emphasis on early detection of the outbreak.

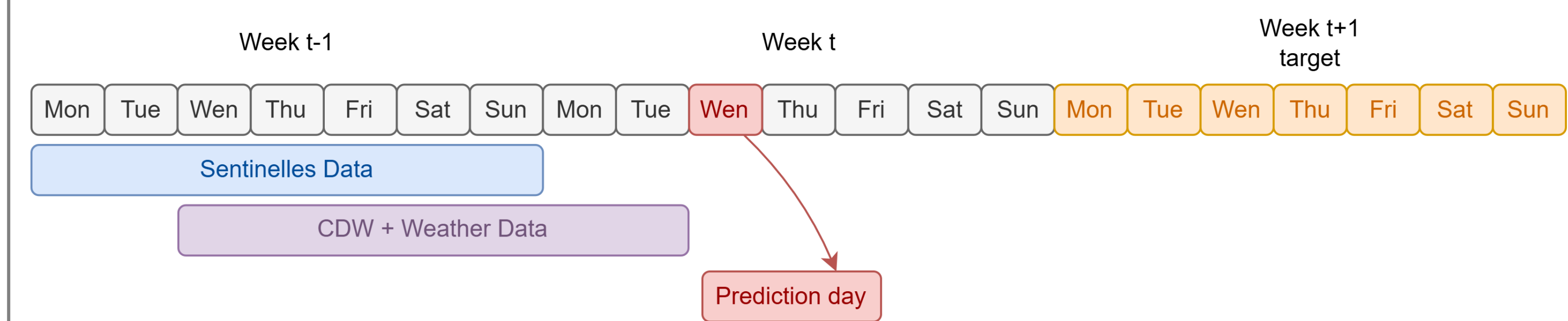
Combining data extraction and predicted models



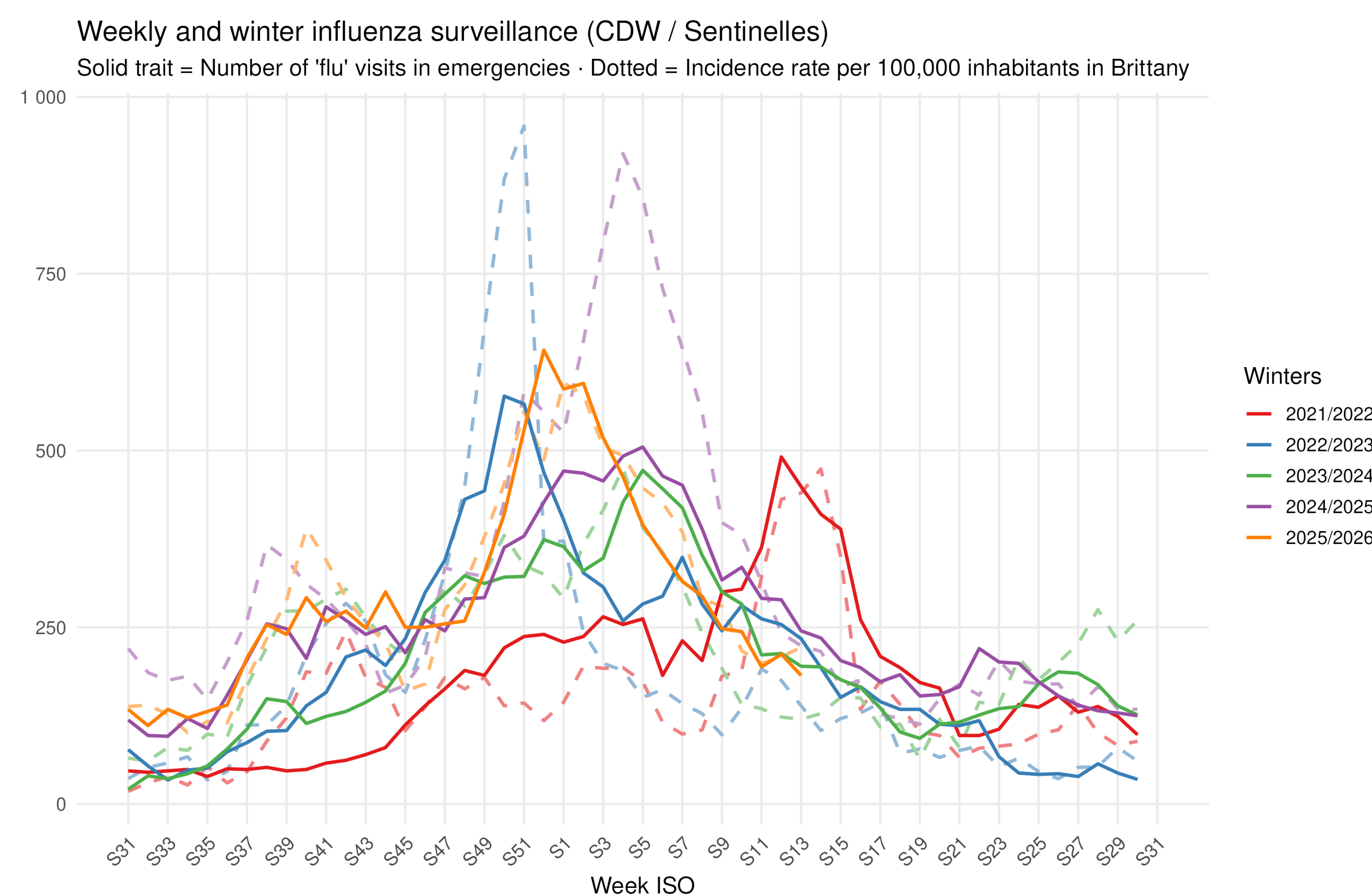
Data extraction



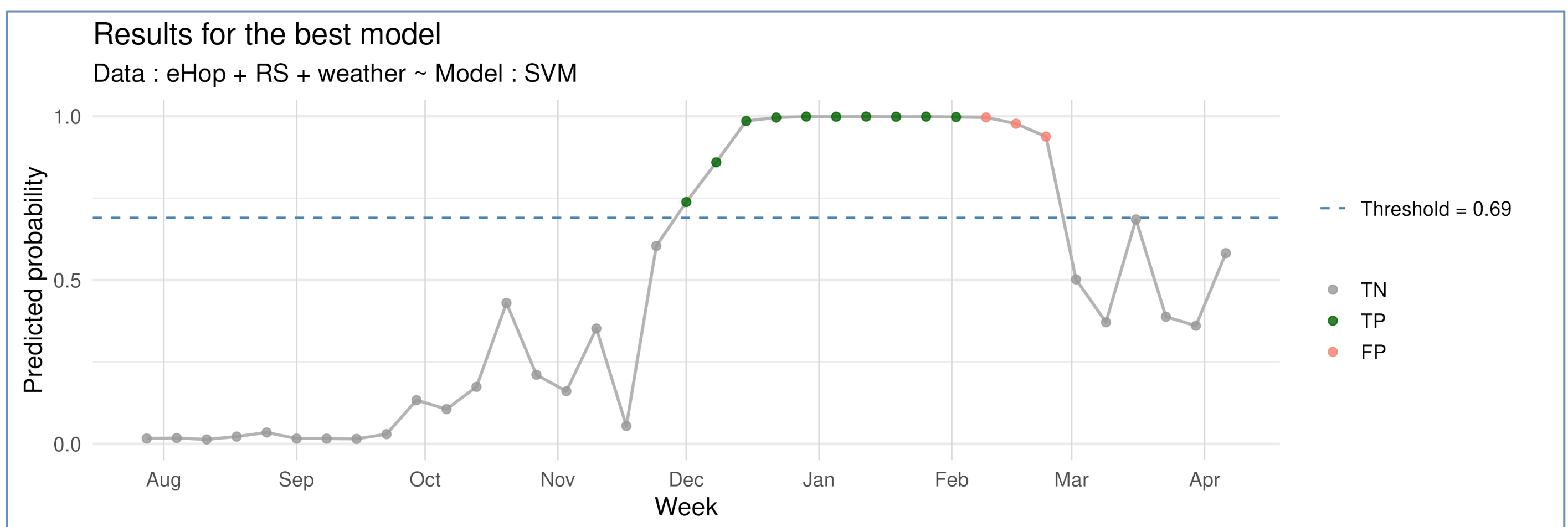
➤ Data used to predict the status of week t+1



Results



Model	TP	FP	TN	FN	Recall	Precision	F1	DeltaStart	DurationDiff
rl_ehop	8	2	25	2	0.8	0.800	0.800	2	0
rl_ehop_weather_red	8	2	25	2	0.8	0.800	0.800	2	0
rf_ehop	8	1	26	2	0.8	0.889	0.842	2	-1
svm_ehop	8	1	26	2	0.8	0.889	0.842	2	-1
svm_ehop_weather_com	10	4	23	0	1.0	0.714	0.833	0	3
svm_ehop_weather_red	10	4	23	0	1.0	0.714	0.833	0	3
svm_ehop_weather_com_RS	10	3	24	0	1.0	0.769	0.870	0	3
svm_ehop_weather_red_RS	10	4	23	0	1.0	0.714	0.833	0	3



Conclusion

- We propose combining heterogeneous data sources through machine learning models to consistently improve forecasting accuracy and responsiveness of epidemiological indicators.
- It is possible to get closer to real time thanks to the CDW, whose data are updated with a delay of day+1. It is even possible to go further by trying to predict the beginning of an epidemic the following week.
- Future work could explore deep learning approaches such as MLPs and perform weight tuning to optimize the contribution of each variable and improve prediction performance.

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GitHub repository with the poster and the authors' contacts